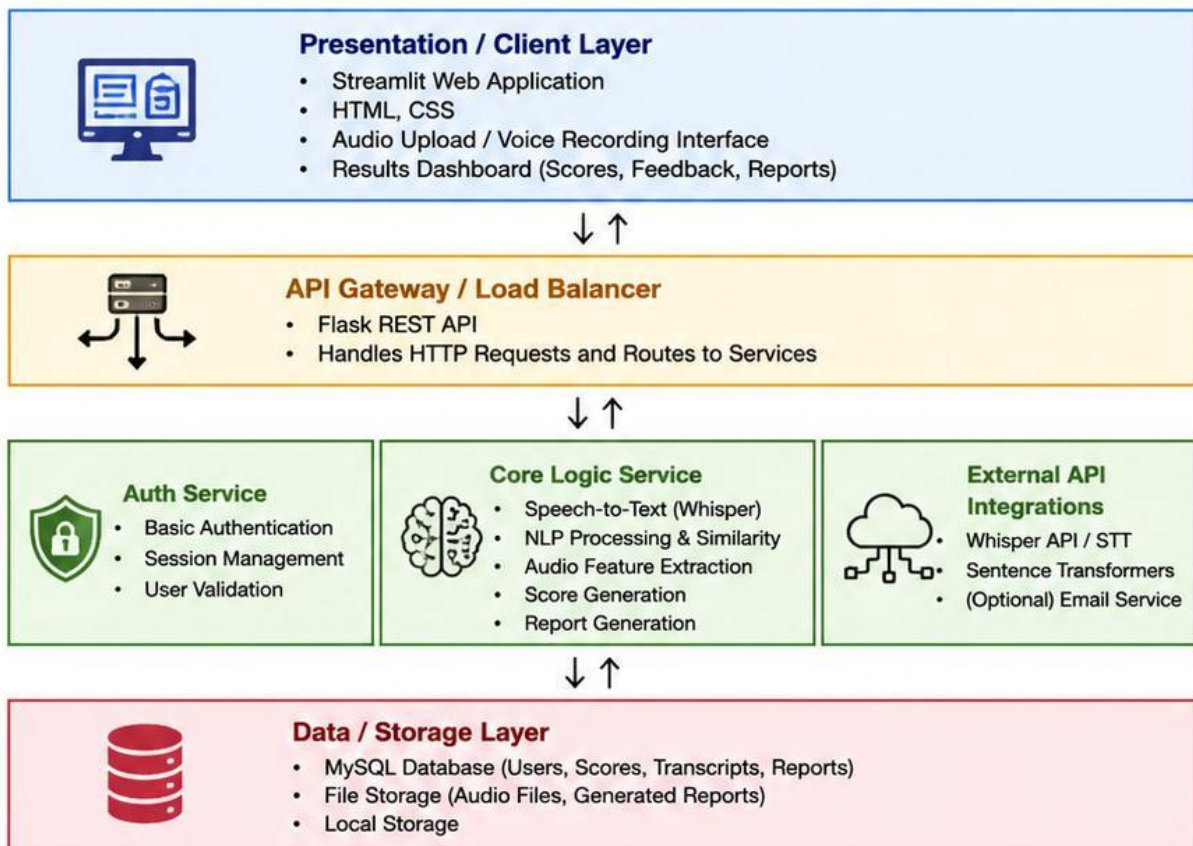


Solution Architecture

Date	26 June 2026
Team ID	N/A
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Provides web UI for uploading audio, viewing transcript, scores, waveform, and downloading report.</i>	<i>Streamlit, HTML, CSS</i>
<i>[API Gateway]</i>	<i>Not used separately because Streamlit handles the app flow directly. Streamlit</i>	<i>Streamlit</i>
<i>[Microservices]</i>	<i>Handles speech-to-text, semantic similarity, audio feature extraction, scoring, and PDF generation.</i>	<i>Python, Whisper, Sentence Transformers, Librosa, Matplotlib, ReportLab</i>
<i>[Database]</i>	<i>Not used currently. Audio/report files are temporarily stored during processing.</i>	<i>Temporary File Storage / Local Storage</i>